NORTHWATCH / 01 HARDWARE & INTEGRATION

CloudGate + Septentrio

Use the mosaic-go X5 for the first unit. The PoC already specifies the X5 and its SBF diagnostics. Bench-check that feed before buying more units. The G5 P6 is a credible lower-cost, compact alternative once its diagnostics and interface pass the same replay and bench checks.

Item	Published specification	Northwatch configuration
CloudGate LTE WW Rev 4 / CG0124-12135	i.MX280 at 450 MHz, 128 MB RAM, 512 MB flash. Guide lists 30 MB application partition, 270 MB free data partition and internal FAT32 microSD.	Local service, bounded logs, lightweight web UI. Confirm writable space, card support and free memory on the actual firmware.
Comms and power	Two 10/100 Mb/s RJ45 ports (WAN and LAN), two expansion slots and 9-33 V DC input. LTE hardware is fitted.	Private LAN cable to indoor helm display. Separate local power by default. PoE via compatible injector and endpoint only after validation. Radios off.
mosaic-go X5 One active GNSS antenna	GPS L1/L2/L5, Galileo E1/E5a/E5b/E6, GLONASS L1/L2/L3, BeiDou B1/B2/B3, QZSS and NavIC. Receiver position output up to 100 Hz.	Outdoor antenna with verified band coverage. Use SBF for RF status, receiver health and position. Verify USB host or supported serial path on CloudGate.
mosaic-go G5 P6 Alternative	Quad-band receiver, dual-antenna heading capability, up to 100 Hz and AIM+ Premium. Kit lists EUR 600 vs EUR 695 for X5.	Evaluate before scaling if smaller size, lower power or the EUR 95 kit saving matters. Confirm the exact kit, firmware and required diagnostics.

Model and budget

Septentrio lists G5 P3, P6 and P8, with no mosaic-G5 P5 in the current module catalogue. If P5 means PolaRx5, that is a separate reference-station receiver. Confirm the part number before ordering.

Core per unit, excluding VAT: about EUR 1,660-2,170. X5 EUR 695, CloudGate EUR 459-476, antenna EUR 199, coax/mount/weatherproofing EUR 75-250, interface/storage EUR 80-200 and enclosure/power/bench lead EUR 150-350. Quote helm display, ship cable and optional PoE separately.

Sources / Septentrio kit prices | model catalogue | X5 | G5 P6 | CloudGate guide | DigiKey pricing. Planning snapshot, 10 Nov 2026.

NORTHWATCH / 02 RUNTIME & TRIAL

What the pilot measures

Run one passive passage. The recorder keeps logging while powered, including during loss of the helm link. The watch can view silent status and record observations. Review alert placement, sound and acknowledgement with the crew afterward.

Function	Planned runtime and output
Acquire and record	Start on boot. Read RFStatus, ReceiverStatus and PVTGeodetic where supported. Log arrival time, event time, state changes and crew observations. Use a monotonic clock for age checks.
Store and export	Rotate within a storage quota. Measure log size per hour with the chosen SBF profile and derive retention from usable capacity. Record write failures. Export approved logs locally.
Show status at helm	Local browser over dedicated Ethernet. Show receiver health, last update age and reason. Expire stale display data on link loss and recover after fresh updates.
Log candidate advisories	Record suspected interference or spoofing, fix unavailable, receiver unavailable and link lost. Show no anomaly detected as receiver status. Keep the first passage silent.

Use cases and KPIs

Trial case	Measure	Interpretation
Normal passive passage	Valid log time / powered time, log gaps, bytes/hour and advisory candidates per watch.	Establish a baseline and size storage for the intended passage.
Receiver or helm cable disconnect	Time to unavailable, time to recover and logging continuity during helm link loss.	Confirm the viewer clears stale status while the powered logger keeps writing.
Shielded RF test or labelled replay	Detection latency from known onset, misses by scenario, false advisories and position error at advisory.	Separate receiver detection in RF tests from UI behaviour in application replay.
Independent fix or optional AIS	Time offset, source, position uncertainty and any radar/AIS mismatch.	Plot reported versus independently observed position when a valid reference exists. AIS is supporting evidence only.

KPIs are measurement definitions. Agree thresholds before the trial. Capture measured power and boot-to-ready time on the assembled kit. Continuous operation depends on approved power and storage; backup runtime is a bench measurement for the chosen configuration.

Guidance / MCA GNSS guidance | IMO onboard AIS guidance.